

베이지안 기계학습

1. Introduction to Bayesian inference

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Orientation (for weeks 1 to 7 classes)

- ▶ Basics of Bayesian inference (Week 1)
- ▶ Markov chain Monte Carlo (MCMC) (Weeks 2 to 3)
- ▶ High-dimensional linear regression models (Weeks 4 to 5)
- ▶ Bayesian network (Week 6)
- ▶ Model selection (Week 7)

Orientation

- ▶ During weeks 2 to 6, we plan to introduce fundamental concepts and practice using R.
- ▶ Bayesian inference using R will mainly rely on the [rstan](#) package.
- ▶ In this class, it will be assumed that students have a general understanding of statistics at a basic level.
- ▶ Additionally, it will be assumed that students can install R and have basic proficiency in its usage.

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- ▶ Basics of Bayesian inference
- ▶ Differences with frequentist approach
- ▶ Bayesian inference for binomial models
- ▶ Remark and next week

Basics of Bayesian inference

Statistical inference

$$Y \sim f(\cdot | \theta)$$

- ▶ $f(\cdot | \theta)$: (parametric) model or the distribution of Y
- ▶ θ : parameter of interest (unknown)
- ▶ Y : random variable (y : realization of Y)
- ▶ Goal: inference about the parameter θ , based on data y .
- ▶ We will focus on **Bayesian inference**.

Three components of Bayesian inference

Bayesian inference: likelihood function

- ▶ A model $f(y | \theta)$ is a distribution of the data y .
 - When Y is a continuous random variable, $f(y | \theta)$ is a probability density function (pdf).
 - When Y is a discrete random variable, $f(y | \theta)$ is a probability mass function (pmf).
- ▶ The **likelihood** function is the same as a pdf or pmf $L(\theta) = f(y | \theta)$, but it is now considered as a function of θ .
- ▶ The likelihood function contains the information about θ included in the data y .

Examples of likelihood functions

- ▶ If $Y \sim B(n, \theta)$, then

$$L(\theta) = f(y | \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}.$$

- ▶ If $Y \sim N(\theta, 1)$, then

$$L(\theta) = f(y | \theta) = (2\pi)^{-1/2} \exp \left\{ -\frac{1}{2}(y - \theta)^2 \right\}.$$

- ▶ If $Y \sim N(\mu, \sigma^2)$, then $\theta = (\mu, \sigma^2)$ and

$$L(\theta) = f(y | \theta) = (2\pi\sigma^2)^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2}(y - \mu)^2 \right\}.$$

Bayesian inference: prior distribution

- ▶ Bayesians express their uncertainty about θ using probability distributions before observing y .
- ▶ A **prior distribution**: $\theta \sim \pi(\theta)$.
- ▶ There is no “true” or “correct” prior.
- ▶ But the choice of priors does affect the results.

Examples of priors

- ▶ If $0 < \theta < 1$ is the success probability in the binomial distribution $B(n, \theta)$, we can use the following priors.
 - $\theta \sim Unif(0, 1)$ (uniform distribution on $[0, 1]$)
 - $\theta \sim Beta(a, b)$ (beta distribution)
- ▶ If $\theta \in \mathbb{R}$ is the mean of the normal distribution $N(\theta, 1)$, we can use the following priors.
 - $\theta \sim N(\mu_0, \sigma_0^2)$
 - $\theta \sim t_m$ (t -distribution with degrees of freedom m)
- ▶ The tuning parameters in the prior are referred to as **hyperparameters**.

Bayesian inference: posterior distribution

- ▶ The **posterior distribution** is the uncertainty distribution of θ after observing the data y .
- ▶ Bayes theorem provides the rule for updating the prior:

$$\pi(\theta | y) = \frac{f(y | \theta)\pi(\theta)}{f(y)} \propto f(y | \theta)\pi(\theta),$$

where $f(y) = \int f(y | \theta)\pi(\theta)d\theta$.

- ▶ In words: Posterior $\propto$ Likelihood $\times$ Prior.
- ▶ The posterior distribution is the conditional distribution of the parameter θ given the data y .

Differences with frequentist approach

Frequentist vs. Bayesian

- ▶ Frequentists view θ as fixed and Y as random.

- (Meaning of probability)

The probability of an event E is

$$\mathbb{P}(E) = \frac{\text{frequency of } E}{\text{the number of repetitions}}.$$

- ▶ Bayesians view y as fixed and θ as random.

- (Meaning of probability)

A probability represents a measure of (personal) uncertainty.

Frequentist vs. Bayesian

- ▶ Bayesians quantify uncertainty about fixed but unknown parameters, θ , by treating them as random variables.
- ▶ Bayesian inference on θ is performed based on its conditional distribution given the data y (posterior distribution).
 - Frequentists use an estimator $\hat{\theta} = \hat{\theta}(Y)$.
- ▶ Bayesians believe that inference on θ should depend only on the data y we have observed.
 - Frequentists consider the data we could observe, but have not yet observed (e.g., confidence intervals).

Review: frequentist estimators

- ▶ Method of moments

$$\widehat{\mathbb{E}}_{\theta}(Y) = \frac{1}{n} \sum_{i=1}^n Y_i,$$

$$\widehat{\text{Var}}_{\theta}(Y) = \frac{1}{n} \sum_{i=1}^n (Y_i - \widehat{\mathbb{E}}_{\theta}(Y))^2$$

- ▶ Maximum likelihood estimator (MLE)

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} L(\theta)$$

Bayesian inference: point estimation

- ▶ Posterior mean

$$\mathbb{E}(\theta \mid y) = \int \theta \pi(\theta \mid y) d\theta$$

- ▶ Maximum a posteriori (MAP)

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \pi(\theta \mid y)$$

- ▶ Posterior median

$$\operatorname{med}(\theta \mid y) = \theta^* \quad \text{satisfying} \quad \pi(\theta \leq \theta^* \mid y) = 1/2.$$

Review: confidence Interval

- ▶ A random interval $(L(Y), U(Y))$ is called a $100(1 - \alpha)\%$ **confidence interval** for θ if

$$\mathbb{P}(L(Y) < \theta < U(Y) \mid \theta) = 1 - \alpha.$$

- ▶ It means that if we repeatedly observe data sets $Y^{(1)}, \dots, Y^{(K)}$,

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{i=1}^K I(L(Y^{(i)}) < \theta < U(Y^{(i)})) = 1 - \alpha.$$

- ▶ A frequentist would NEVER say “the probability of the $100(1 - \alpha)\%$ confidence interval containing θ is $1 - \alpha$ ”.

Bayesian inference: interval estimation

- ▶ For a given data y , an interval $(L(y), U(y))$ is called a $100(1 - \alpha)\%$ **credible interval** for θ if

$$\pi(L(y) < \theta < U(y) \mid y) = 1 - \alpha.$$

- ▶ The posterior probability that θ lies in $(L(y), U(y))$ is $1 - \alpha$.
- ▶ Bayesians can say “the probability of the $100(1 - \alpha)\%$ credible interval containing θ is $1 - \alpha$ ”.
- ▶ We can find multiple $(L(y), U(y))$, i.e., a credible interval may not be unique.

Highest posterior density (HPD) set

- ▶ An $100(1 - \alpha)\%$ **HPD set** for θ is an interval $(L^*(y), U^*(y))$ defined by

$$(L^*(y), U^*(y)) = \{\theta : \pi(\theta | y) \geq k\},$$

where k is the largest number such that

$$\int_{\theta: \pi(\theta|y) \geq k} \pi(\theta | x) d\theta = 1 - \alpha.$$

- ▶ The HPD set will be an interval when the posterior is unimodal.
- ▶ If the posterior is multimodal, the HPD set might be a discontinuous set.

Advantages of Bayesian inference

- ▶ Bayesian concepts are arguably easier to interpret than frequentist ideas.
- ▶ Excellent at quantifying uncertainty in complex problems (e.g., missing data, deep neural network, etc.).
- ▶ Bayesians can incorporate scientific knowledge via the prior.
- ▶ Provides a framework to incorporate data/information from multiple sources.

Disadvantages of Bayesian inference

- ▶ Less common/familiar.
- ▶ Picking a prior is subjective.
- ▶ Computing can be slow or unstable for complex problems.
- ▶ Nonparametric methods are challenging.

Bayesian inference for binomial models

Binomial model

- ▶ Suppose we observe

$$X_1, \dots, X_n \mid \theta \stackrel{iid}{\sim} \text{Ber}(\theta). \quad (1)$$

- ▶ $X_i \in \{0, 1\}$ for all $i = 1, \dots, n$, where $X_i = 1$ means that a success was observed in the i th trial.
- ▶ $0 < \theta < 1$ is the “success probability”.
- ▶ Let $Y = \sum_{i=1}^n X_i$, then

$$Y \mid \theta \sim B(n, \theta). \quad (2)$$

- ▶ Models (1) and (2) are equivalent.

Binomial model

$$Y \mid \theta \sim B(n, \theta)$$

- ▶ The pmf is $f(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$, $y = 0, 1, \dots, n$.
- ▶ For example, we observe n patients & Y recover.
- ▶ Let θ be the probability of a patient recovering from cancer.
- ▶ We wish to estimate θ via the Bayesian approach.

Binomial model

$$Y \mid \theta \sim B(n, \theta),$$
$$\theta \sim \text{Beta}(a, b)$$

- ▶ A commonly used prior is the beta prior.
- ▶ The pdf is $\pi(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$, $0 < \theta < 1$.
- ▶ For example, we can use $\text{Beta}(1, 1) = \text{Unif}(0, 1)$ prior for θ .

Binomial model

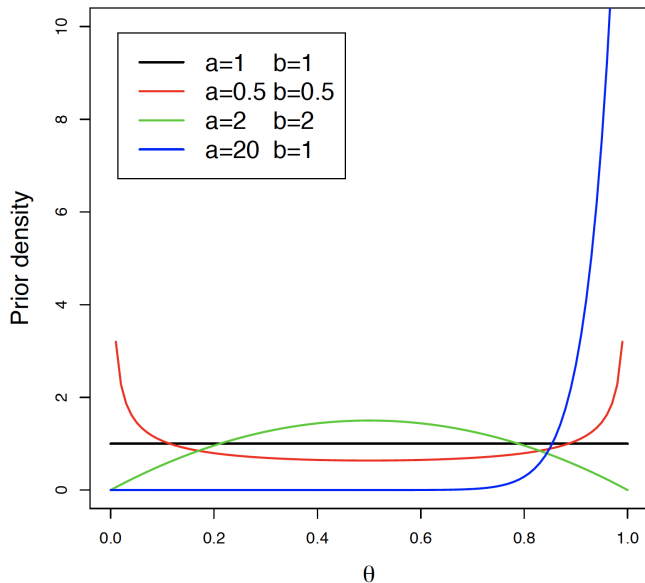
$$Y \mid \theta \sim B(n, \theta),$$

$$\theta \sim \text{Beta}(a, b),$$

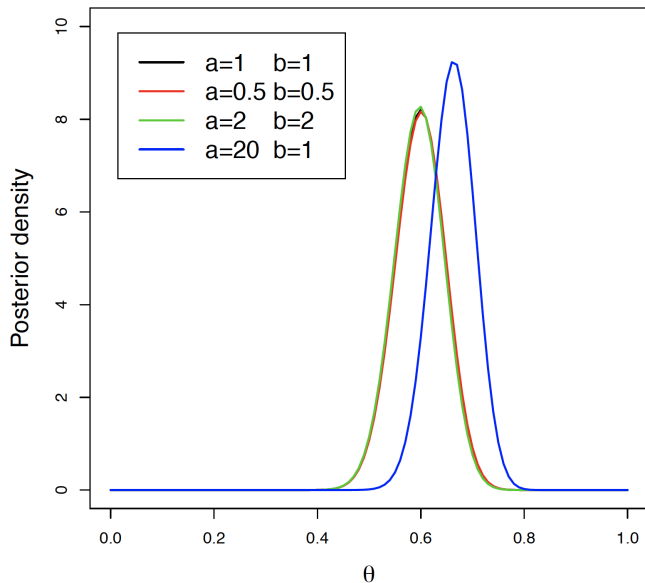
$$\theta \mid Y \sim \text{Beta}(Y + a, n - Y + b)$$

- ▶ It can be shown that the posterior is also beta distribution.
- ▶ Prior and posterior belong to the same distribution family: the prior is **conjugate** for the likelihood.

Plot of different beta priors



Plot of the corresponding posteriors



Posterior computation

- ▶ Let's calculate the posterior distribution:

$$\begin{aligned}\pi(\theta | y) &= \frac{f(y | \theta)\pi(\theta)}{f(y)} \propto f(y | \theta)\pi(\theta) \\ &\propto \theta^y (1 - \theta)^{n-y} \theta^{a-1} (1 - \theta)^{b-1} \\ &\propto \theta^{y+a-1} (1 - \theta)^{n-y+b-1}.\end{aligned}$$

- ▶ The last expression is proportional to $Beta(y + a, n - y + b)$'s pdf, thus $\pi(\theta | y)$ should coincide with $Beta(y + a, n - y + b)$'s pdf.

Summarizing the posterior

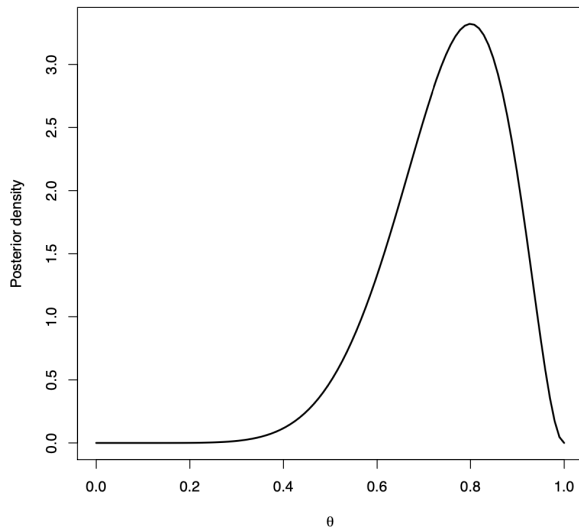


Figure: $\pi(\theta | y)$ when $y = 8$, $n = 10$ and $a = b = 1$.

Summarizing the posterior

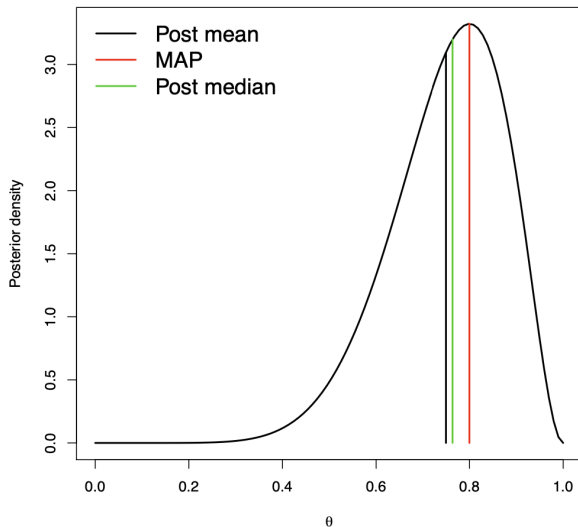


Figure: $\pi(\theta | y)$ when $y = 8$, $n = 10$ and $a = b = 1$.

Summarizing the posterior

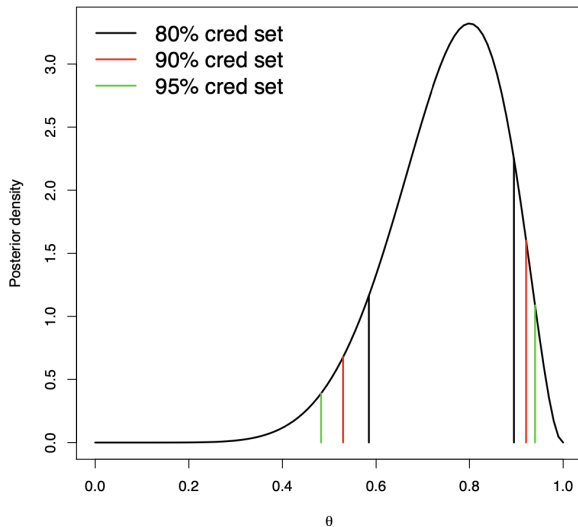


Figure: $\pi(\theta | y)$ when $y = 8$, $n = 10$ and $a = b = 1$.

Estimation and prediction

- ▶ We will take a closer look at the methods for point estimation and prediction.
- ▶ Suppose we have an observation y from $B(n, \theta)$.
- ▶ We are interested in the estimation of θ .
- ▶ Among point estimation methods, let's focus on the posterior mean.

Point estimation

- ▶ Let $\hat{\theta} = \mathbb{E}(\theta \mid y)$ be the posterior mean.
- ▶ The mean of $Beta(y + a, n - y + b)$ (posterior) is

$$\hat{\theta} = \frac{y + a}{y + a + n - y + b} = \frac{y + a}{a + b + n}.$$

- ▶ We can decompose it into

$$\hat{\theta} = \frac{n}{a + b + n} \underbrace{\left(\frac{y}{n}\right)}_{\text{sample mean}} + \frac{a + b}{a + b + n} \underbrace{\left(\frac{a}{a + b}\right)}_{\text{prior mean}}.$$

Point estimation: interpretation

$$\hat{\theta} = \frac{n}{a+b+n} \underbrace{\left(\frac{y}{n}\right)}_{\text{sample mean}} + \frac{a+b}{a+b+n} \underbrace{\left(\frac{a}{a+b}\right)}_{\text{prior mean}}$$

- ▶ The “Bayes estimator” $\hat{\theta}$ is a weighted average of the usual frequentist estimator (sample mean) and the prior mean.
- ▶ As n increases, the sample mean part is weighted more heavily. In fact, $\hat{\theta} \approx y/n$ for all large n .
- ▶ In general, as the sample size increases, the likelihood dominates the prior.

Point estimation: choice of hyperparameters

$$\hat{\theta} = \frac{n}{a+b+n} \underbrace{\left(\frac{y}{n}\right)}_{\text{sample mean}} + \frac{a+b}{a+b+n} \underbrace{\left(\frac{a}{a+b}\right)}_{\text{prior mean}}$$

- ▶ $\frac{n}{a+b+n}$ and $\frac{a+b}{a+b+n}$ are relative weights of n and $a+b$.
- ▶ We call $a+b$ the **prior sample size**.
- ▶ The choice of (a, b)
 - $a+b$: prior sample size
 - $\frac{a}{a+b}$: prior guess of θ

Point estimation: advanced

- ▶ Using Bayesian inference, we can naturally infer not only the parameter θ , but also any arbitrary function of θ .
- ▶ Suppose we are interested in the log odds,

$$\eta = \log\left(\frac{\theta}{1-\theta}\right).$$

- ▶ It might be difficult to calculate the posterior of η directly.

Point estimation: advanced

- ▶ In this case, we can use the Monte Carlo (MC) approximation:

$$\begin{aligned} I &= \mathbb{E}\{h(X)\} = \int h(x)f(x)dx \\ &\approx \frac{1}{N} \sum_{i=1}^N h(X_i), \quad X_1, \dots, X_N \stackrel{iid}{\sim} f. \end{aligned}$$

- ▶ MC is a useful tool for summarizing a posterior.
- ▶ It may not be particularly useful in univariate cases, but in more challenging problems, it is the most effective approach.
- ▶ In our case, X and f correspond to θ and $\pi(\theta | y)$, respectively.

Point estimation: advanced

- ▶ Returning to our example, let's assume we are interested in the log odds $\eta = \log\left(\frac{\theta}{1-\theta}\right) = h(\theta)$.
- ▶ Suppose we have M posterior samples,

$$\theta^{(m)} \stackrel{iid}{\sim} \pi(\theta \mid y), \quad m = 1, \dots, M.$$

- ▶ Then, we can use an MC approximation based on

$$\eta^{(m)} = \log\left(\frac{\theta^{(m)}}{1 - \theta^{(m)}}\right), \quad m = 1, \dots, M.$$

Prediction 1

- ▶ **Prediction** refers to the inference of future data values based on already observed data, $x = (x_1, \dots, x_n)$.
- ▶ Suppose we are interested in the prediction of X_{n+1} .
- ▶ Frequentists may perform predictions as follows:
 1. Calculate an estimate $\hat{\theta}$ based on x .
 2. Use $\mathbb{P}(X_{n+1} = 1 \mid \hat{\theta})$ as a prediction probability.
- ▶ However, this method struggles to incorporate the uncertainty in $\hat{\theta}$.

Prediction 1

- ▶ Bayesians use $\mathbb{P}(X_{n+1} = 1 \mid x)$ as a prediction probability of X_{n+1} based on the data x .
- ▶ Note that

$$\begin{aligned}\mathbb{P}(X_{n+1} = 1 \mid x) &= \int_0^1 \mathbb{P}(X_{n+1} = 1 \mid \theta, x) \pi(\theta \mid x) d\theta \\ &= \int_0^1 \mathbb{P}(X_{n+1} = 1 \mid \theta) \pi(\theta \mid x) d\theta \\ &= \int_0^1 \theta \pi(\theta \mid x) d\theta \\ &= \mathbb{E}(\theta \mid x) = \frac{a + \sum_{i=1}^n x_i}{a + b + n}.\end{aligned}$$

Prediction 2

- ▶ Now, suppose we are interested in the prediction of $Z = X_{n+1} + \cdots + X_{n+m}$.
- ▶ This is equivalent to predicting how many successes will be observed in m future trials.
- ▶ Again, frequentists may use

$$\mathbb{P}(Z = z \mid \hat{\theta}) = g(z \mid \hat{\theta}), \quad z = 0, 1, \dots, m,$$

as a prediction density, where $g(z \mid \theta)$ is the pmf of $B(m, \theta)$.

Prediction 2

- Bayesians use the following prediction density:

$$\begin{aligned}\mathbb{P}(Z = z \mid x) &= \int_0^1 \mathbb{P}(Z = z \mid \theta) \pi(\theta \mid x) d\theta \\ &= \int_0^1 \binom{m}{z} \theta^z (1 - \theta)^{m-z} \pi(\theta \mid x) d\theta \\ &= \binom{m}{z} \frac{\Gamma(a + b + n)}{\Gamma(a + \sum_{i=1}^n x_i) \Gamma(b + n - \sum_{i=1}^n x_i)} \\ &\times \frac{\Gamma(a + \sum_{i=1}^n x_i + z) \Gamma(b + n - \sum_{i=1}^n x_i + m - z)}{\Gamma(a + b + n + m)},\end{aligned}$$

for $z = 0, 1, \dots, m$.

- The above prediction distribution is called the Beta-Binomial distribution.

Remark and next week

Remark

- ▶ In simple models including the binomial model, the posterior $\pi(\theta | y)$ can be calculated in a closed form.
- ▶ Unfortunately, in complex models, the posterior $\pi(\theta | y)$ is not available in a closed form, in other words,

$$f(y) = \int f(\theta | y)\pi(\theta)d\theta \quad \text{is difficult to compute.}$$

- ▶ In general, we need computational methods to obtain (approximate) samples from the posterior $\pi(\theta | y)$:
 $\Rightarrow$ Markov chain Monte Carlo (MCMC) methods.

Next week

Markov chain Monte Carlo (MCMC) I